A systematic review of COVID-19 diagnostics uptake pathways in Low- Middle Income and High-Income countries							
Author(s)	Title of the article	Journal name	Type of article	Month & Year Published	Country	Income category (LMIC or HIC)	How COVID-19 testing was increased
Kanchan Mukherjee	Integrating technology, innovation and policy: COVID-19 and HTA	Elesevier- Health policy and technologuy	research article	Feb-21	India	LMIC	1)Department of science and technology launched initiatives to support COVID-19 response. This lead to development of the FELUDA test for COVID-19 diagnosis. 2)Deployed HTA to assess the innovations 3)testing strategy that used RDP (antigen) and RT-PCR for negative RDT
Pravash Budhathoki, Dhan Bahadur Shrestha, Sitaram Khadka, Santoshi Giri	COVID-19 Status in Nepal and the Way Forward	Europasian Journal of Medical Sciences	review article	Jun-20	Nepal	LMIC	1)Introduction of RDTs in 3 districts 2)PCR testing expanded from national public health lab to 17 centres
Atieme Ogbolosingha	COVID-19 pandemic: Review of impediments to public health measures in SSA	American Journal of Preventive Medicine and Public Health · January 2020	review article	Jun-20	South Africa	UMIC	1) Private pathology labs in SA to supplement public facilities 2) Mobile testing units in SA
					Nigeria	LMIC	1) Mobile testing units
Zheng Li , Cynthia Jones, Girum S. Ejigu, Nisha George, et al	Countries with delayed COVID-19 introduction – characteristics, drivers, gaps and opportunities	Globalisation and Health	research article	2021	24 countries	Mixed	1)Lesotho, Sao Tome & Principle, Cook Islands, Kiribati, Marshal Islands, Micronesia, Nauru, Palua,Samoa, Solomon Islands no incountry testing so export COVID testing was done

Olivier Vandenberg, Delphine Martiny, Olivier Rochas , Alex van Belkum, Zisis Kozlakidis	Considerations for diagnostic COVID-19 tests	Nature reviews-Microbiology	review article	Oct-20	Several		1)Leveraging platforms used for SARS-CoV and MERS to develop tests 2) government intervention seg financial investments and policies instituted to support manufacturiong capacity 3) USA introduced legislation to alter regulatory frameworks governing lab developed tests 3)continued operation of manufacturing businesese 4) repurposing of manufacturing capacity 5)Centralised system in UK, clinical lab network in Germany, 6) setting up networks of regional sequencing centres hosted in academic and or publi chealth agencies,,
Shabir Ahmad Lone and Aijaz Ahmad	COVID-19 pandemic – an African perspective	Emerging Microbes & Infections	review article	May-20	Several	mixed	1) Continental efforts like AFCOR hel[ped increase testing labs in African from 2 in Feb to 43 by March
Amy S Patteerson and Emmanuel Balogun	African responses to COVID-19: The reckoning of agency	african: studies review	review article	Feb-21	Several		1) Donations of test kit from China 2) Contenental efforts eg Africa CDC developed a COVID-19 Joint Continental Strategy which inlcuded training HCWs in lab dx, 3) PACT formed to stregthen testing capacity on the continent, it procured testing kits and distributed them to several countries, and also created distribution warehouses & distribution hubs 4)AFCOR created btn Africa CDC, WAHO and WHO to increase number og lad conducting testing 5) Partnered with Unitaid, gates foundation, CHAI on devt of rapid diagnostics
Alfonso Medinilla, Bruce Byiers and Philomena Apiko	Africa regional responses to COVID-19	ecdpm making policies work	discussion paper	May-20	Several		1) central role played by Africa CDC in boosting testing capabilities and trainings (it improved testing capacity of labs across the continet, supplying essential equipment to 48 countries and also provided training 2) Throuf PACT, estbalished warehousing and distribution hubs across Africa, coordinated the pooled procurment of diagnostics material, and training CHWs to support contact tracing

Zachary Desson, Emmi Weller, Peter McMeekin, Mehdi Ammi	An analysis of the policy responses to the COVID-19 pandemic in France, Belgium, and Canada	Health policy and technology	review article	Dec-20	Canada	HIC	1) Initially had one testing centre and gradually capacity was built in other provinces
					France	HIC	1) France massive testing of suspected cases by hospitals, labs, vet clinics and in both govt and private facilities, which led to non harmonised reporting and later special temporary regislation allowed a secure paltform to allow consolidation of results,
					Belgium	HIC	
Luke Taylor	Uruguay is winning against covid-19. This is how	BMJ	feature	Sep-20	Uruguay	HIC	1)Uruguay ramped testing by developing a national testing kit to shield it from market shortages 2) created a network of testing labs in weeks; which included academic labs 3) decentralised testing
Elisabeth Mahase	Covid-19: mass testing in Slovakia may have helped cut infections	BMJ	news	Dec-20	Slovakia	HIC	population wide (mass testing) using an rapid antigen test administered by trained medical peronnel instead of self testinf. Onluy those that tested and were negative were allowed to move
GRID COVID-19 Study Group	Combating the COVID-19 pandemic in a resource-constrained setting: insights from initial response in India	BMJ Global health	Practice article	Oct-20	India	LMIC	1) Increased testing and collection centres 2) expanded testing criteria, 3) engagement of private labs, and existing labs upgraded 4) free of costs testing 5) sample collection kiosks and mobile COVID testing vans 6)(ma6) made in india testing kits on market
Thuong Vu Nguyen, Quang Dai Tran, Lan Trong Phan, Long Ngoc Vu	In the interest of public safety: rapid response to the COVID-19 epidemic in Vietnam	BMJ Global health	Practice article	Jan-21	Vietnam	LMIC	1)collecting samples from people at risk 2) local production of RT-PCR diagnostic kit 3)developed and stocked diagnostic testing products (primers & probes) and antibody tests 4) staff training and assistance 5) increased lab capacity from 2 to 120 labs in 8 months

Senjuti Saha, Madhukar Pai	Can COVID-19 innovations and systems help low- and middle-income countries to re-imagine healthcare delivery?	Med	Commentary	Apr-21	Bangladesh	LMIC	1) Opened up multiple testing labs 2)multiple groups of people within any basic lab training asked to volunteer & support lab techs 3) govt & lab offered training on sample collection across the nation 4) local reagent & supply vendors solved supply chain puzzles to import supplies from different parts 5) expanded program of immunisation and local WHO office leveraged expertise & network of distributing vaccines to instead transport C-19 samples 6) local sequencing capacity and did sequencing for C-19
					India	LMIC	1) utilisation of existing technologies & resources and repurposing existing technologies
Sangchun Choi, Chul Han, Jisook Lee, Sang-Il Kim, In Byung Kim	Innovative screening tests for COVID-19 in South Korea	Clinical and Experimental Emergency Medicine	Commentary	Apr-20	South Korea	HIC	1) established drive through testing for C-19, 2) testing for C-19 at a walking-through site
Patrick Mbulaje , Steven Munharo, Symon F. Nayupe, Parth Patel	Challenges of COVID-19 testing in low-middle income countries (LMICs): the case of Malawi	Journal of laboratory and precision medicine	letter to the editor	Sep-20	Malawi	LIC	1) with support from UNICEF an existing RT-PCR machine at Lilingwe NRL was set up 2) complentary efforts by teaching and research instiutions and Malawi-Liverpool Wellcome Trust 3) increased the number of testing facilities in the country 4)
Jean B. Nachega, Ashraf Grimwood, Hassan Mahomed, Geoffrey Fatti, et al	From Easing Lockdowns to Scaling Up Community-based Coronavirus Disease 2019 Screening, Testing, and Contact Tracing in Africa—	Clinical infectious disease	view point	Jan-21	South Africa	UMIC	1) deployed CHCWs for screening & referral for testing 2) leveraging UBV infrastructure for COVID-19, and other diseases like malaria, TB,HIV
					DRC	LIC	1) mplementation of multiprofessional health system of nurses, drs, pharmacists, medical students &CHCWS in senitisation, screening, and testing referral
					Rwanda	LIC	1) Sample collection and referral for testing

	Shared Approaches, Innovations, and Challenges to Minimize Morbidity and Mortality						
Nancy Odendaal	Constructing an "infrastructure of care" – understanding the institutional remnants and socio-technical practices that constitute South Africa's Covid-19 response	Urban geography	view point	Jul-20	SA	UMIC	1) community wide testing using CHWs in areas identified thru a social vulnerability index
H. Sibiri, S.M. Zankawah, D. Prah	Coronavirus diseases 2019 (COVID-19) response: Highlights of Ghana's scientific and technological innovativeness and breakthroughs	ethics medicine and public health	commentary	May-20	Ghana	LMIC	1) pooling techinique 2) dron etechnology to transport samples 3)COVID-19 tracker up to identify who needs to be tested 4)development of RDTs
Jimoh Amzat, Kafayat Aminu, Victor I. Kolo, Maryann C. Danjibo	Coronavirus outbreak in Nigeria: Burden and socio-medical response during the first 100 days	internationa ljournal of infectious diseases		Sep-20	Nigeria	LMIC	1) prioritised those that need to be tested-returnees, symptomatic, contacts, those in high prevalence areas 2) trainingexperts in RT-PCR by Africa CDC 3) Increased number of testing labs from 5 to 23 within 6 months - only public minus private were testing
A. Waris, U.K. Atta, M. Ali, A. Asmat, A. Baset	COVID-19 outbreak: current scenario of Pakistan	new microbes and new infections	mini review	May 202	Pakistan	LMIC	1)Increased number of testing labs 2) national institute of health opened up a mobile testing lab 3) NIH partnered with national disaster management authority to launch training

							program for paramdecis and lab staff to overcome HR issues
Ryenchindorj Erkhembayar, Emma Dickinson, Darmaa Badarch, Indermohan Narula et al	Early policy actions and emergency response to the COVID- 19 pandemic in Mongolia: experiences and challenges	The Lancet Global Health	health policy	Sep-20	Mongolia	LMIC	1)ordered for probes, primers and possitve controls from Japan 2) used already available platforms 3) lab testing organised by national influenza centre virology lab 4) random community samopliong and walk in testing
Adugna Abera, Habtamu Belaya, Aboma Zewudea, Bokretsion Gideya, et al	Establishment of COVID-19 testing laboratory in resource-limited settings: challenges and prospects reported from Ethiopia	GLOBAL HEALTH ACTION	narrative	Oct-20	Ethiopia	LIC	1)Led by the EPHI, the malaria and NTD research lab used platforms originally used for other diseases eg ifluenza& shcistosomisais to set up the 1st testing lab 2) Training of lab personnel by the incountry virology experts
Santosh Dhakal *and Surendra Karki*	Early Epidemiological Features of COVID-19 in Nepal and Public Health Response	frontiers in medicine	original research	Aug-20	Nepal	LMIC	1)increased number of testing labs from 1 to 20 in a short period 2) included veterinary labs
Saeed Anwar, Mohammad Nasrullah and Mohammad Jakir Hosen	COVID-19 and Bangladesh: Challenges and How to Address Them	frontiers in medicine	policy brief	Apr-20	Bangladesh	LMIC	1) locally developed rapid diagnostic 2) expansion of labs from one to 17 including academic institution

Ahsan Nawaz, Xing Su, Muhammad Qasim Barkat, Sana Asghar, et al.	Epidemic Spread and Its Management Through Governance and Leadership Response Influencing the Arising Challenges Around COVID-19 in Pakistan—A Lesson Learnt for Low Income Countries With Limited Resource	frontiers in medicine	narrative review	Dec-20	Pakistan	LMIC	1) increased labs precoid to 57 labs after covid 2) training program for staff to increase number ofo techs and experts in molecular biology
Nega Assefa, Jemal Yousuf Hassen, Desalegn Admassu, Mussie Brhane et al	Existing Facilities in Public Health Emergency Management	frontiers in medicine	case study	Jun-21	Ethiopa	LIC	1) support and gyidance from EPHI to academic institutions to set up testing in the academic institution. Support inlcuded training, protocols among others
Percy Herrera-Añazco, Angela Uyen-Cateriano, Edward Mezones-Holguin, Alvaro Taype-Rondan et al.	Some lessons that Peru did not learn before the second wave of COVID-19	international journal of health management	letter to the editor	2021	Peru	UMIC	1)deployed massive use of rapid serological tests as diagnostic and epidemiological tools
Jaroslav Frnda and Marek Durica	On Pilot Massive COVID-19 Testing by Antigen Tests in Europe. Case Study: Slovakia	infectious disease reports	research paper	Jan-21	Slovakia	HIC	Population/country wide rapid antigen testing

Azizah F. Siddiqui, Manuel Wiederkehr, Liudmila Rozanova and Antoine Flahault	Situation of India in the COVID-19 Pandemic: India's Initial Pandemic Experience	International journal of Environmental research and public health	case report	Dec-20	India	LMIC	1)gradual increase in number of testing labs from 1 to 11 to 885 by end of June 2) private labs included too
Ki Tae Kwon, Jae-Hoon Ko, Heejun Shin, Minki Sung and Jin Yong Kim	Drive-Through Screening Center for COVID-19: a Safe and Efficient Screening System against Massive Community Outbreak	infectious diseases, microbiology and parasititology	brief communication	Mar-20	Korea	HIC	1)introduction of drive through screening and testig centres
Tafadzwa Dzinamarira, Mathias Dzobo, Itai Chitungo	COVID-19: A perspective on Africa's capacity and response	journal of medical virology	review	Jun-20	Nigeria	LMIC	1) Leveraged the existing health infrastructure programs used for previous poliovirus
					Sierra Leone	LIC	1) In Sierra Leone they leveraged structures previoulsy deployed for ebola
Ghulam N Dhabaan, Walid A Al-Soneidor, Nezar N Al-Hebshi	Challenges to testing covd-19 in conflict zones: Yemen as an example	journal of global health	view point	Jun-20	Yemen	LIC	1)used 4 of the 6 central public health labs to test, but this was largely supported by WHO (who trained and also gave supplies)
Al-Musthapha AI, Tijani AA, Oyewo M, Ibrahim A, Elelu N, et al	Nigeria's race to zero COVID-19 cases: True disease burden or testing failure?	Journal of Global Health	view point	Aug-21	Nigeria	LMIC	1) increased labs from the initial 6 to 124 in 4 months 2) leveraged the exisiting platforms used for endemic and sporadic outbreaks 3) deployed xpert platform previoulsy used for HIV and TB 4) private and corporate labs included in testing
Alberto Boretti	The Increasing Number of Infected in Victoria,	integrative journal of medical sciences	case report	Aug-20	Australia	HIC	1)identification of hotspots and mass testing including door to door testing

	Australia since June 15, 2020, is the Result of Over-Testing and Over-Controlling without Safety						
Godfred Acheampong, Michael Owusu, Bernard Nkrumah, Priscilla Obeng-Boadi, et al.	Laboratory capacity in COVID-19 diagnosis and the need to enhance molecular testing in Ghana	global security: health, science and policy	research article	Mar-21	Ghana	LMIC	1)initially used 3 of the 4 public health labs that had capcity and also used 2 of the acadmic institutions that had capacity 2)gradual increase in number of labs including non clinical labs eg vet labs and academic labs 3) accredited private labs 4)leveraged the GeneXpert platforms 5) scientists among first to sequence the sars-cov-2
Karen E. Ferree, Adam S. Harris, Boniface Dulani, Kristen Kao, et al.	Stigma, Trust, and procedural integrity: Covid-19 testing in Malawi	world development	research article	Jan-21	Malawi	LIC	1) increased labs from 1 to 11 in around 2.5 months, 2)focus on suspected cases , immediate contacts and travelers into the country
María Alejandra Benítez, Carolina Velasco, Ana Rita Sequeira, Josefa Henríquez et al.	Responses to COVID-19 in five Latin American countries	Health policy and technology	narrative review	Aug-20	Chile	HIC	2) Adopted pooled testing in elderly homes 2) introduced low cost tests for COVID
					Peru	UMIC	1)they increased labs doing testing gradually
Peter J. Chipimo, Danielle T. Barradas, Nkomba Kayeyi, Paul M. Zulu, et al.	First 100 Persons with COVID-19 — Zambia, March 18–April 28, 2020	morbidity and mortality weekly report	report	Oct-20	Zambia	LMIC	1)expanded national testin gstrategy to include asymptomatic persons with possibel COVID-19 exposures and no international travel
Tinashe K. Nyazika, Rabelani Kaela, Mathias Mugoni, Kudakwashe	Implementation of Antibody Rapid Diagnostic Testing versus Real-Time	American society for microbiology	perspective	Jul-20	Ghana	LMIC	1) Deployed sample pooling later in the pandemic
					Malawi	LIC	1) Increased testing coverage by increasing number of testing labs

Musomekwa, et al.	Reverse Transcription-PCR Sample Pooling in the Screening of COVID-19: a Case of Different Testing Strategies in Africa				South Africa	UMIC	1) Deployed targeted community symptom screening and testing for both symptomatic and asymptomatic individuals in some provinces
					Zimbabwe	LMIC	1) ntroduced RDT as a screening test strategy to complement RT-PCR
Bruce J. Tromberg, Tara A. Schwetz, Eliseo J. Pérez-Stable, Richard J. Hodes, et al.	Rapid Scaling Up of Covid-19 Diagnostic Testing in the United States — The NIH RADx Initiative	New England journal of medicine	special report	Sep-20	USA	HIC	1)Initiatives like RADx to sup port the development, production scale-up, and deployment of accurate, rapid tests across the country
Oppah Kuguyo, Andre Pascal Kengne, and Collet Dandara	Singapore COVID-19 Pandemic Response as a Successful Model Framework for Low-Resource Health Care Settings in Africa?	Integrative biology	Narrative	2020	Singapore	HIC	1)Drew lessons from the sars-cov-1 outbreak in 2002 2) in country biotechnology companies produced several tests
Adisu Kebede, Betty Lanyero, Berhane Beyene, Mayur Lalji Mandalia, et al.	Expanding molecular diagnostic capacity for COVID-19 in Ethiopia: operational implications, challenges and lessons learnt	PanAfrican medical journal	Perspective	Jan-21	Ethiopia	LIC	1)identification and assesssment ot public and private labs with support from EPHI, WHO etc 2) on iste bench training, onsite mentorship and drills for the lab personnel 3) calibration of existing PCR machines to be able to do covid testing using kits donated 4) progressive expansion of network by adding more labs 5) used the national influenza, arbovirus and VHF reference lab at EPHI to estbalish covid testing in the country 6) inclusion of research, academic and private labs in testing

Matthew M Kavanagh, Ngozi A Erondu, Oyewale Tomori, Victor J Dzau,et al.	Access to life-saving medical resources for African countries:	Lancet	View point		Several		1)Africa CDC established pooled purchasing platform
Paul Adepoju	Africa's struggle with inadequate COVID-19 testing	Lancet microbe	News	May-20	Ghana, Nigeria, South Africa	LMIC	4) Drone delivery service stsarted to transport samples,
					Nigeria	LMIC	1)Involved already existing viral hemorrhagic fevers lab network & other labs with molecular diagnostic capacity to increase coverage 2) health start up LifeBank established covid-19 mobile testing centres
					South Africa	UMIC	2) Involved the private sector in testing
Pascale Ondoa, Yenew Kebede, Marguerite Massinga Loembe, Jinal N Bhiman, et al.	COVID-19 testing in Africa: lessons learnt	Lancet microbe	Comment	Jul-20	Several		1)continental efforts coordinated by AFTCOR included teaming up with SA national institute for ID, Senegalese Institute Pasteur Daker & WAHO to train expert staff from refe labs in Africa.2)Africa CDC Pathogen genomic intelligence institute part of joint ocntinental covid-19 startegy to boost sequecning capcity of 16 AU member states 3)Nigeria activated covid-19 testing at 26 sites using HIV molecular testing and TB xpert instruments 4)Ethiopia reconfigured exisiting paltforms 5)Ethiopia involved academic and animal health labs 6) PACT formed to increase testing and organising AU states as on elarge customer to be able to compete
Lincoln Leehang Lau, Natalee Hung and Kendall Wilson	COVID-19 response strategies: considering inequalities	International journal for equity in health	Commentary	Aug-20	Phillipines	LMIC	1)reconfiguring TB testing facilities to meet testing demands

	between and within countries						
Olivier Vandenberg, Delphine Martiny, Olivier Rochas, Alex van Belkum *and Zisis Kozlakidis*	Considerations for diagnostic COVID-19 tests	Nature reviews Microbiology	review	Mar-21	USA	HIC	1)introduction of legislation to alter the regulatory framework governin glab developed tests, 2) active scouting and import og resources outside usual territories, 3) repurpoing manufacturing capacity
Edward Kim	Drawing on Israel's Experience Organizing Volunteers to Operationalize Drive-Through Coronavirus Testing Centers	Disaster Medicine and Public Health Preparedness	field report	2020	Israel	HIC	1)operationalised drive through coronavirus testinf centres
Lauran K. Evans, Austin Shinagawa, Sarah Sutton, Lisa Calvo	COVID-19 Drive-Through Point of Screening and Testing (POST) System: A Safe, Efficient, and Adaptable Model for Nasopharyngeal Swab Collection	Disaster Medicine and Public Health Preparedness	original research	2020	USA	HIC	1)implementation of covid-19 drive through point of screening and testing
Anthony Kwabena Sarfo, Shankar Karuppannan	Application of Geospatial Technologies in the COVID-19 Fight of Ghana	Indian National Academy of Engineering	original research	Jul-20	Ghana	LMIC	1)deployed drone technology to help with sample transport

David Lee and Jaehong Lee	Testing on the move: South Korea's rapid response to the COVID-19 pandemic	Transportation Research Interdisciplinary Perspectives			South Korea	HIC	1)drive through testing 2)used IT to do surveillance and contact tracing . IT inlcuded mobile phone tracking, credit card transaction and video footage from surveillance cams
Dean T. Jamison, Kin Bing Wu	The East–West Divide in Response to COVID-19	Transportation Research Interdisciplinary Perspectives	perspective	Jun-21	Several (China, USA, Vietnam, Europe		1)commisoning of private biotech firms and academic isntitutions to undertake genomic analysis that lead to identificatiomn of the genome hence allowing rapid development of of PCR tests 2)Germany and Korea among first countries to develop tests once genome was shared by China 3)Vietnamm developed itw own kits
PatriciaJ.Garcia, AlexAlarco´n, AngelaBayer, PauloBuss, et al.	COVID-19 Response in Latin America	American journal of tropical medicine and hygiene	perspective piece	Sep-20	Chile	HIC	1) lab screening capacity expanded by including all the country's institutions with PCR equipment and supplying reagents to the lab network of universities and institutes
					Peru	UMIC	1) government purchased serologic tests which were validated and swiftly deployed to scale up testing
Ramanan Laxminarayan, Shahid Jameel, and Swarup Sarkar	India's Battle against COVID-19: Progress and Challenges	American journal of tropical medicine and hygiene	editorial	Aug-20	India	LMIC	1) India ramped up testing by rolling out rapid antigen testing 7 months into the pandemic
Thinley Dorji	The Gross National Happiness Framework and the Health System Response to the COVID-19 Pandemic in Bhutan	American journal of tropical medicine and hygiene	perspective piece	Dec-20	Bhutan	LMIC	1) expanded RT-PCR testing from main centre to more gepgrpahically spread centres 2) increased human resource available (reapproproation of human resource) including engagment of students 3)free covid testing for its citizens 4) initiated population testing of all persons older than 10 years

Mathew Alexander, Lynn Unruh, Andriy Koval and William Belanger	United States response to the COVID-19 pandemic, January–November 2020	Health Economics, Policy and Law	article	Jan-21	USA	HIC	1)authorisation of private and state and local health labs to develop tetst in February (1st commercial test approved in March -3 months after 1st case). 2) enlistment of private labs in testing.
Aditya Shah, Douglas Challener, Aaron J. Tande, Maryam Mahmood et al.	Drive-Through Testing: A Unique, Efficient Method of Collecting Large Volume of Specimens During the SARS-CoV-2 (COVID-19) Pandemic	Foundation for Medical Education and Research	brief report	Jul-20	USA	HIC	1)drive through testing at Mayo Clinic
Marguerite Massinga Loembé, Akhona Tshangela, Stephanie J. Salyer, Jay K. Varma, Ahmed E. Ogwell Ouma and John N. Nkengasong	COVID-19 in Africa: the spread and response	Nature medicine	comment	Jul-20	Several		1) Africa CDC efforts to increase no of labs 2) leveraged network of community health agents used for previosu diseases eg polio 3) Pasteur institute Dakar Senegal evalusated RDTs
Marc Mendelson, Shabir Madhi,	South Africa's coronavirus testing strategy is broken and not fit for purpose: It's time for a change	South African Medical Journal	Guest editorial	Jun-20	South Africa	UMIC	1) Initiated the community screening and testing (CST) program 2) Recruited field workers to support CST
Reports							

OECD/European Union 2020	How resilient have European health systems been to the COVID-19 crisis?	Health at a Glance: Europe 2020		Jun-20	Several		1) France limited testing to patients with serious conditions at start of pandemic 2) Iceland conducted large scale testing regime early in outbreak 3) Iceland later did voluntary self referral testing regardless of status 4) Lithuania set up mobile points for testing
Janice Thompson and Sinead McMurray	COVID-19:Testing for Sars-CoV-2 in the UK; and the Use of Testing and Contact Tracing in Selected Countries	Northern Ireland assembly		May-20	UK	HIC	1)Early on "containment phase" testing was for the symptomatics after travel and contacts 2) Later moved to delay phase and tested only those requiring admission to hospital 3) Later rolled out to symptomatic NHS staff and their households 4)academic insittuitons involved
Elizabeth Rough	Coronavirus: Testing for Covid-19	House of commons library	Briefing paper	Oct-20	UK	HIC	1)Decentralised appraoch of testing 2) Inclusion of both public and private providers 3) Inclusion of academic and research institutions 4) containment phase 5) delay phose 6) expanded eligibility to those above 65 with symptoms and later to anyone with 5 yrs and above with symptoms 7) developed point of care tests
Department of Health and Social Care	Coronavirus (COVID-19) Scaling up our testing programmes		Report	Apr-20	UK	HIC	1) new testing lans set up 2) inlcuded universities and reseatch institutes 3) support to new innovators of tests

Ministry of Finance Planning and Economic development	COVID-19 Interventions Report	MoFPED		Oct-20	Uganda	LIC	1)Operationalising testing labs at border points 2) Accreditation of UVRI bu Africa CDC and WHO 3) Used Genexpert at border points
Infectious diseases sociery of America	COVID-19 Prioritization of Diagnostic Testing	Infectious diseases sociery of America		Mar-20	USA	HIC	1) Expanded testing to a wider group of symptomatic patients in March 2020
FDA	South Korea's Response to COVID-19	Center for Devices and Radiological Health			Korea	HIC	1) Korea leveraged structures from previous epidemics like mers, 2) govt designed EUA pathway for tests and 1st test on market in Jan 3)invited private providers and assured them of ready market 4) linked manufactirers with clinical bodies to conduct clinical studies reducing
Nigeria Centre for Disease Control	National strategy to scal eup access to C-19 disease testing in Nigeria			2020	Nigeria		1) leveraged existing disease structures for Malaria, TB and HIV, 2) built capcity by increasing labs 3) developed a national strategy to scal eup access to c-19testing
Mathieu Ductatel and Francois Godement	Fighting COVID-19: East Asian Responses to the pandemic		Policy paper	Apr-20	North Korea	HIC	1) Started developing very quickly the NAA tests 2) Also started developing serological tests 3) Fast approval processfor medical devices including tests 3) 4) mass testing 7) offering PCR tests offered free of charge

					China	HIC	1) started developing very quickly the NAA tests 2) Also started developing serological tests 3) Fast approval processfor medical devices including tests
					Singapore	HIC	1) PP partnerships to develop/produce tests
Congressional Research service	COVID-19 testing: Key issues		In focus	Apr-20	USA	HIC	1) initially used public health labs but later included commercial and clinicial labs in hospitals and universities, 2) FDA allowed labs and commercial manufacturers ti use and market test kits prior to receiving EUA
Koichi Kameda, Mady Malheiros Barbeitas, Rosangela Caetano, Ilana Lowy et al.	Testing COVID-19 in Brazil:fragmented efforts and challenges to expand diagnostic capcity at the Brazilian Unified Nationak Health system	Reports in Public health	bried communication	Feb-21	Brazil	UMIC	1)initially testing in hospital testing for symptomatics 2) Included public labs and labs from public universities 2) leveraged HIV/AIDs viral load testing labs 3) increased labs testing for covid 4) public private partnerships 5) private developed leveraged molecualr system developed for blood screening to test covid 6) fast tracking market authorisation of COVID tets by EUA
Pan African Health Organisation	Pan American Health Organisation Response to COVID-19 in the Americas		report	May-20	Sevral		1) ordered and provided test kits to member states 2) conducted trainings 3) leveraged the regions nationak influenza centred and severe acute respiratory infection lab network